EXP 3

Aim

To design and simulate a 10 GHz microwave directional coupler with 10 dB coupling and a two-way 3 dB power divider using 50 Ω ports, analyze their S-parameters over the frequency range of 8–12 GHz, and optimize the coupling-section length by 5% to compare coupling, isolation, directivity, insertion loss, and power-division ratio.

Software required:

MATLAB

Program:

```
clc;
clear;
close all;

% Simulation parameters
f0 = 10e9;                    % Center frequency: 10 GHz
f = linspace(8e9,12e9,401);
Z0 = 50;                      % Port impedance
c = 3e8;
lambda0 = c/f0;

% Design specifications
C_dB = 10;                    % Directional coupler coupling
D_dB = 3;                     % Power divider output
L = lambda0/4;                % Initial coupling-section length
L_opt = 1.05*L;               % 5% length optimization

% Coupling coefficient
k = 10^(-C_dB/20);

% Frequency-dependent phase
beta = 2*pi*f/c;
theta = beta*L;
theta_opt = beta*L_opt;

% Directional coupler model
S11 = zeros(size(f));
S21 = cos(theta);
S31 = 1i*k.*sin(theta);
S41 = zeros(size(f));
```

```
% Ideal two-way 3 dB power divider
S21_div = ones(size(f))/sqrt(2);
S31_div = ones(size(f))/sqrt(2);

% Plot directional coupler S-parameters
figure;
plot(f/1e9,S11_dB,'k','LineWidth',1.5);
hold on;
plot(f/1e9,S21_dB,'b','LineWidth',1.5);
plot(f/1e9,S31_dB,'r','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Directional Coupler S-Parameters');
legend('S11','S21','S31','Location','best');

% Plot before and after optimization
figure;
plot(f/1e9,S31_dB,'b','LineWidth',1.5);
hold on;
plot(f/1e9,S31_opt_dB,'r--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Coupled Output (dB)');
title('Coupling Before and After 5% Length Optimization');
legend('Before Optimization','After Optimization');

% Plot ideal power divider
figure;
plot(f/1e9,20*log10(abs(S21_div)),'b','LineWidth',1.5);
hold on;
plot(f/1e9,20*log10(abs(S31_div)),'r--','LineWidth',1.5);
grid on;
```

```
% Optimized directional coupler
S21_opt = cos(theta_opt);
S31_opt = 1i*k.*sin(theta_opt);

% Convert S-parameters to dB
S11_dB = 20*log10(abs(S11)+1e-12);
S21_dB = 20*log10(abs(S21)+1e-12);
S31_dB = 20*log10(abs(S31)+1e-12);

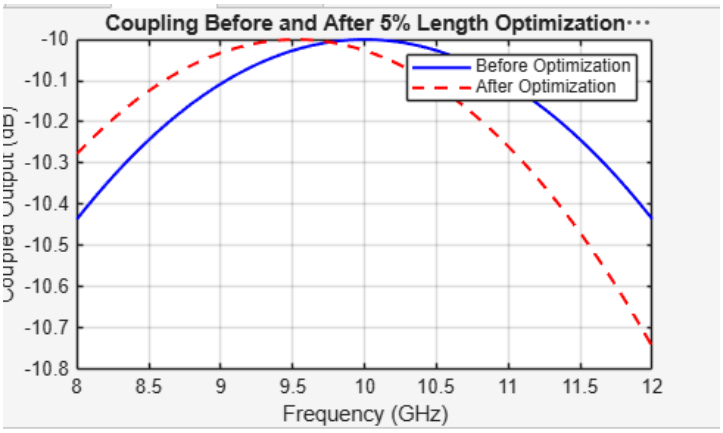
S21_opt_dB =
20*log10(abs(S21_opt)+1e-12);
S31_opt_dB =
20*log10(abs(S31_opt)+1e-12);
```

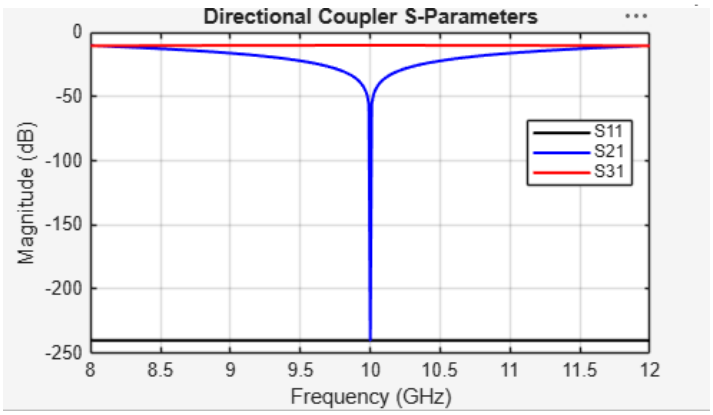
```
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Two-Way 3 dB Power Divider');
legend('S21','S31','Location','best');

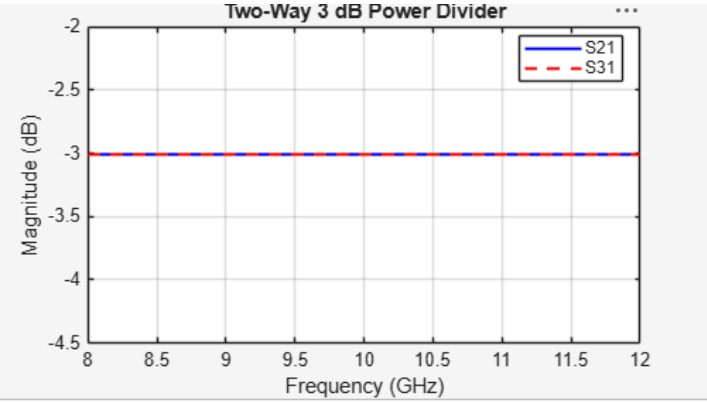
% Display design parameters
fprintf('Center frequency: %.2f
GHz\n',f0/1e9);
fprintf('Port impedance: %.0f Ohms\n',Z0);
fprintf('Initial length: %.2f mm\n',L*1e3);
fprintf('Optimized length: %.2f
mm\n',L_opt*1e3);
fprintf('Ideal divider output: %.2f dB\n', ...
    20*log10(1/sqrt(2)));
```

Output:

```
Port impedance: 50 Ohms
Initial length: 7.50 mm
Optimized length: 7.88 mm
Ideal divider output: -3.01 dB
```






Two-Way 3 dB Power Divider
S21
S31
Magnitude (dB)
-2
-2.5
-3
-3.5
-4
-4.5
8
8.5
9
9.5
10
10.5
11
11.5
12
Frequency (GHz)